

Understanding and Analyzing the Performance Scalability of N-body Problem

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ABSTRACT

The rapid data growing recently years has provided us much more information about the universe and has contributed to the rising needs of strong computer powers in the meantime as it means that it is not feasible to handle such large amount of data without strong computer powers. Therefore, it is important to understand how the computer and programs work to help model and solve the complex problems, so that we can discover and explore more principles other than just numbers. This paper mainly focuses on the N-Body problem which is a common problem that can be extends to many fields, aiming to find the subsequent motion of multiple objects with known initial position, velocity, and mass in classical mechanics. The paper discusses about the understanding of both brute force and Barnes-Hut algorithm and analyses the performance and scalability of two approaches on the N-Body problem under a personal computer environment. In addition to this, the serial simulation and the parallel simulation of the brute force implementation are also get compared in the perspective of running time and the scalability in the paper.

CCS CONCEPTS

• Computer systems organization;

KEYWORDS

N-body, brute force method, Barnes-Hut algorithm, scalability, time complexity, galaxy simulation, computational

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1 INTRODUCTION

The N-Body problem is one of the complicated problems that the computer can model. One of the classic usages of N-Body problem is to explore theories and model the structures about the universe. The understanding and analyzing the scalability performance of N-Body problem in this paper is also based on this galaxy simulation.

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In the process of simulation, the computer helps us to model the whole space and discover the interforce that stars interacting with one another, and thus predict and estimate the particles' mass, locations, and movements over time. By doing this, we can have a better understanding of the how galaxy evolves and acts with the power of the computers.

Both brute force and Barnes-Hut algorithm can perform the simulation and solve the N-Body problem, however, the time complexity and the scalability of the two methods are significantly different. It is important to recognize how the performance would change with the behaviours and properties of the inputs given to the algorithms so that we better seed up the computing performance and maximize the use of computer power to help us model and discover complex phenomena which is infeasible to calculate or explore merely by hands. The reasons of such time complexity scalability difference have been discussed in the paper as well.

2 CHALLENGES FACING

The N-Body problem is always not easy to model, so as the galaxy simulation. The main challenges faced in the simulation performing is due to the fact that there are various irregular and dynamically changing behaviors in the galaxy, and the major computational difficulties faced in the simulation process is to deal with the influence which the N bodies in the system interact with one another over time.

3 BRUTE FORCE

To solve the problem and perform the simulation of the galaxy phenomena, a most obvious way is what's known as the brute force method by directly integrating the method with the Newtonian equations of gravitational force. In the brute force approach, all pairs of individual particles are considered, while the total force that act on each particle is the combination of the forces that applied all the other particles.

$$f = \sum_{i \neq n}^N \left(\frac{Gm_n m_i}{r_{ni}^2} \right) \hat{r}_{ni}. \quad (1)$$

Eq. (1) is the Newtonian gravitational force equation, where G is gravitational constant, m is the mass of a particular particle, r is the region of characteristic size, n is the number of particles.

Unfortunately, although this brute force algorithm can calculate out the approximately accurate results, the main disadvantage of this method is its asymptotic computational time as all pair of interactions are computed between all particles. The time complexity of this algorithm scales to $O(N^2)$ where N represent the number of particles in the simulation system (deducing from the actual formula of $\frac{N*(N-1)}{2}$). This complexity implies that the asymptotic computation time increases fourfold as the number of particles

in the system doubles, which makes the algorithm infeasible to perform for a large N as the time consumed by the method grows proportionally.

4 BARNES-HUT

To improve this non-linear time complexity problem, Barnes-Hut algorithm is introduced. Barnes-Hut approximation is an efficient hierarchical algorithm that approximates the forces acting on every body and the particles' movements over time. The algorithm not only produce the accurate result but also accelerates the computation time and enables the simulation to be scalable simultaneously. It is effective and efficient in the way that the algorithm really speeds up the calculation time in the simulation process and only takes $O(N \log N)$ time complexity compared to the naive brute force algorithm which would take $O(N^2)$.

The process of simulating a galaxy using the Barnes-hut algorithm is split into hundreds of time steps, each of which calculates the resultant forces on each particle and updating that particle's position and other properties. The improvement achieved by the Barnes-Hut approximation is mainly by decreasing the number of force calculations necessary per time step as the magnitude of interaction between particles fall off rapidly while distance between them increases. The three-dimensional simulation volume is usually divided up into an octree, so that only particles from nearby cells need to be treated separately, and particles in distant cells can be regarded as a single large particle located in the center of the mass of the cell.

Thus, if a group of bodies is distant away from the body whose net force, we are interesting to compute, the effect of that group of bodies can be approximated by a single equivalent body. Consequently, the algorithm implies that the more distant particles are, the larger possibilities that the group can be approximated by a single large particle, as they become significantly less important to the current calculating body's subsequent motions.

Therefore, this approach helps us to greatly reduce the number of pair interactions of particles that we need to calculate. An oct-tree structure is deployed by the Barnes-Hut algorithm to implement the approach the tree is used to represent a three-dimensional space that depicts the whole galaxies.

The tree built is initially empty only have a root cell which represent a space cell that containing all bodies in the system. Then the bodies in the system are added to the initial empty root cell gradually. When there are more than a fixed number of bodies contained in one cell, the cell would subdivide in to eight equally sized children cells. Thus, this tree structure is able to handle both low and high particle densities spaces, and the resulting oct tree would have internal nodes that represents spaces cells, and the leaves nodes depicts the individual body. The Figure 1 below depicts how a small two-dimensional space can be mapped to its corresponding quadtree.

Once the tree is generated, we need to calculate the net force on every body. In the process of calculating the resultant force on a particular body, nodes in the tree are traversed starting at the root. If the centre of mass of an internal node is far enough away from the body, the bodies that contained in that internal cell and the sub-nodes below that are regarded as a single particle whose

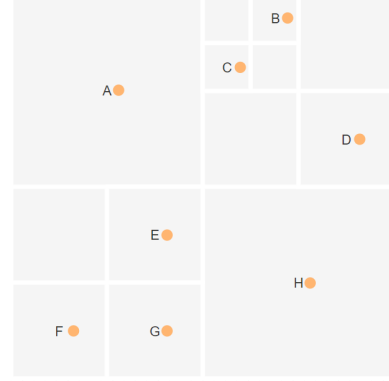


Figure 1: A small two-dimensional space

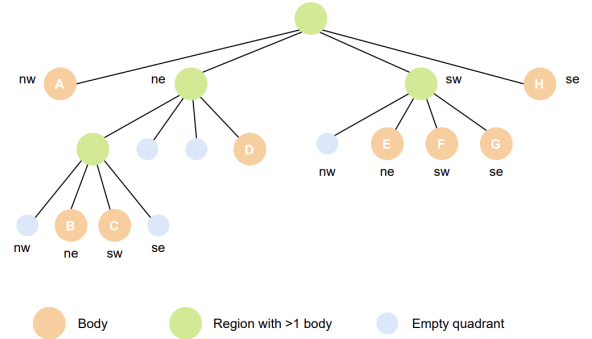


Figure 2: The corresponding quadtree

position and mass are the centroid and total mass of the inner nodes respectively, while the rest of the subtree is not traversed. If the internal nodes are close enough to the body, the cell is perceived as opened and the process is repeated for each of its child's nodes.

The same approach is recursively applying to the cells until reach to the point that a node only containing one child, so that after the process has done, all the bodies motion and location would be updated as well as other properties. In addition to this, the Barnes-Hut approximation method also defines a systematic way to determine whether the node is sufficiently far away from the body or not. The degree of the level of the distance is measured based on the following equation.

$$r = \frac{l}{d} \quad (2)$$

l is the width of the region, d is the distance between the body and the node's center of mass.

The node is perceived as sufficiently far away when this ratio is calculated to be smaller than a threshold value θ . The parameter θ is a regulation parameter between the accuracy of the simulation and the length of the simulation time. The larger values of θ , the quicker the simulation is, whereas the accuracy of the simulation would decrease. Once the $\theta = 0$, no internal node would be treated

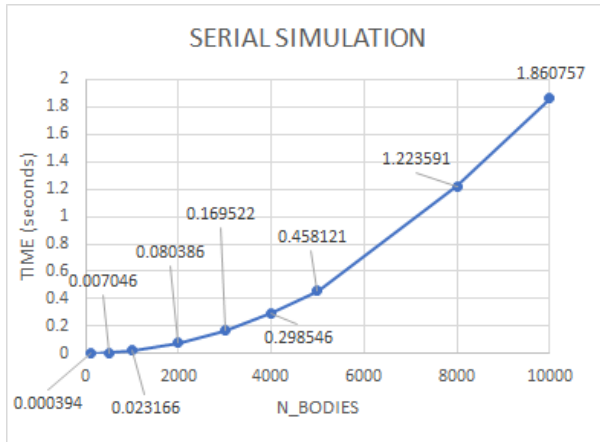


Figure 3: Serial simulation of brute force in one process

as a single body and the algorithm would just simply degrade to a brute force algorithm.

Since the depth of the tree is expected to be $O(\log n)$ and the number of bodies traversed in the tree is in N , so the expected time complexity of the algorithm is $O(n \log n)$. Therefore, compared to a naïve brute force approach, Barnes-Hut algorithm does improve the time consumed to solve the problem a lot, and the time difference between these two methods is substantial.

5 COMPARISON

The figure below (Figure 3) is used to further analyze the scalability of the serial simulation of the brute force method in the case of galaxy simulation. The serial simulation uses only one process and are performed to model the systems that contain different number bodies from small to large. From the figure (Figure 3), the computational time grows proportionally with the time increases further demonstrates the theoretic time complexity of brute force algorithm that was shown as $O(N^2)$.

The figures (Figure 4 and Figure 5) below compare the performance of parallel simulation using different number of processes under systems with different number of particles to see if multiple processes can help speed up the program or not. The parallel simulation partitions the networks into subnetworks and assigns logical processes to those that can be executed using different processes concurrently. Figure 4 and 5 demonstrate the using of more processes in the simulation program can improve the computational costs as expected, which makes it be a more flexible and applicable strategy to reduce the execution time and increases the scalability of the program simultaneously.

From two figures above, we can clearly find out that the parallel simulation is way better than the serial simulation in the terms of computational time consumed. In addition to this, unlike the serial approach, the time consumed by the parallel simulation with multiple processes is not growing substantially when the number of bodies grows rapidly, which further reveals that the parallel simulation with multiple processes is much more capable to scale to a larger problem.

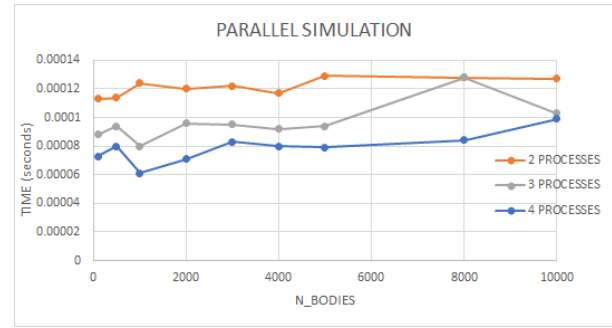


Figure 4: Parallel simulation of brute force with multi-process

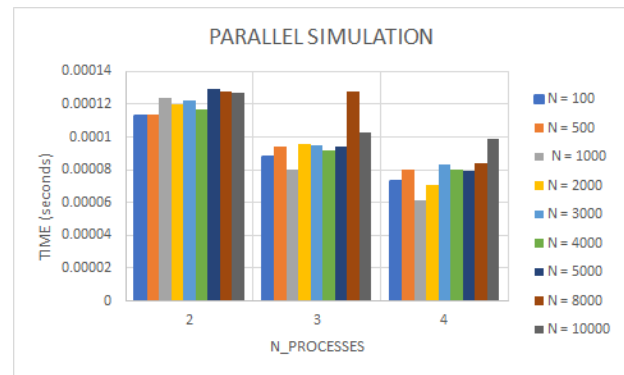


Figure 5: Parallel simulation of brute force with multi-process

To further demonstrate the improvement in the scalability made by the Barnes-Hut approach, the number of force calculations that performance by the Barnes-Hut approach per time step is compared with the number of force calculations that performance in the brute force approach per time step in the figure below (Figure 6). The figure depicts the number of forces calculates made in Barnes-Hut approximation simulation grows almost linear whereas the number of force calculations made in brute force grows substantially. The data once more confirmed our theoretical expectation, as major improvement made by the Barnes-Hut from brute force method is to reduce the number of force calculation per time step.

Figure 5 evidences that Barnes-Hut now has much better and strong scalability than the naïve brute force method, and the Barnes-Hut algorithm can help us to save significantly large amount of computation time now.

With a low number of bodies, a slight streamlined brute force is more efficient than a Barnes-Hut algorithm, but as the N increases, the efficiency increases dramatically, roughly 50% gain for $N < 500$, 80% for $N < 1000$, reaching up to 90% with Larger N s.

6 CONCLUSION

Such efficient algorithm is not only beneficial for the galaxy simulation, but also similar phenomena related to the N body problem, for example, to model and examine the connections between people

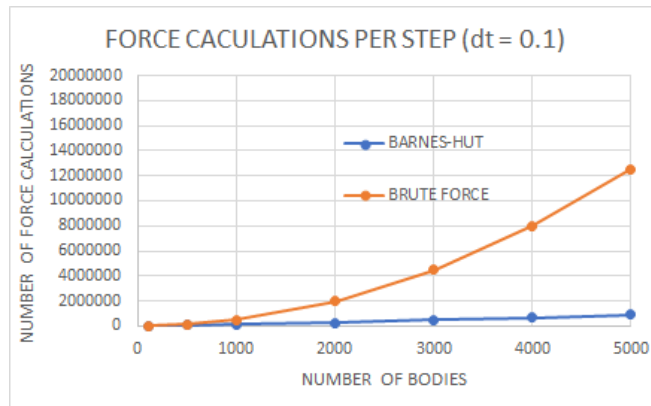


Figure 6: Barnes-Hut approximation simulations

in the social network. The galaxy simulation and social network simulation may seem irrelevant in the first sight, but they do share common properties and need in some degrees. They both require computing forces between pairwise interactions among set of nodes, in the case of astronomical simulation, the force among stars are

the gravitational forces among particles, whereas in the case of network visualization, we try to model the relationship between everyone as the forces that interacting between them.

In conclusion, computer nowadays can be used as exciting tools to model and discover complex phenomena which is infeasible to calculate or explore merely by hands. The difference in the scalabilities of the serial and parallel brute force simulation, and the difference in the scalabilities of the brute force and the Barnes-Hut algorithm both demonstrate the effectiveness and benefits that an efficient and scalable algorithm can bring to us and allow us to do more kind of things

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